

Karim ElSedfy

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EDUCATION

Bowdoin College

B.A. in Computer Science and Mathematics

May 2028

Brunswick, ME

- **GPA:** 3.85/4.00
- **Clubs and Organizations:** ColorStack, Men's Club Rugby
- **Relevant Coursework:** Algorithms, Computer Systems, Artificial Intelligence, Computational Complexity Theory, Data Structures I & II, Deep Learning, Linear Algebra.

EXPERIENCE

Dropbox

Software Engineer Intern

May 2026 – Present

Remote - San Francisco, CA

- Build conversion features on the Teams Formation team that nudge business-domain users on free plans toward paid Team plans, reworking onboarding UI and upgrade flows to drive plan adoption and recurring revenue.
- Develop full-stack changes across a React/TypeScript frontend and Python backend services, leveraging internal gRPC-based RPC and A/B experimentation tooling to ship and measure targeted upsell experiences.

Palmera Systems Co.

Software Engineer Intern

Jun. 2025 – Aug. 2025

Cairo, Egypt

- Architected the face-recognition and anti-spoofing feature for smartIN app (8K daily users), handling dataset curation and model fine-tuning with DeepFace/OpenCV—achieving 96% detection accuracy.
- Engineered robust full-stack web interfaces in Angular (e.g., a responsive school website) and implemented .NET Core back-end services to manage users, content publishing, and online admissions.
- Integrated .NET Core APIs with PostgreSQL to match live captures against database in less than 2s, reducing authentication latency by 50%.

Ungated Research, Bowdoin College

Software Engineer

Mar. 2026 – Present

Brunswick, ME

- Co-maintain a multithreaded Python ETL pipeline (SQLAlchemy, MariaDB, AWS S3) ingesting economics papers from 11+ journals, enriched with Gemini summaries, tags, and translations.
- Engineered error-detection and automated-repair diagnostics to flag and fix articles with missing or invalid fields, and ran Crossref API polling across 10 years of publications to surface 6,080 new articles for ingestion.

Computer Science Department, Bowdoin College

Teaching Assistant

Jan. 2025 – May 2026

Brunswick, ME

- Held weekly office hours (~4 hrs/week) across three semesters for 82 students in Data Structures I & II and Artificial Intelligence, clarifying core concepts in algorithms, data structures, and OOP.
- Delivered targeted feedback on coding assignments, coaching students through debugging, optimization, and algorithmic problem-solving strategies.

PROJECTS

Proteus: Virtual Try-On

[GitHub](#) | [GitHub](#) | [Live Demo](#)

React, Vite, Node.js, Express, WebSocket, Redis, PostgreSQL, Python, PyTorch, Kubernetes (LKE), S3

- Built an event-driven, GPU-accelerated virtual try-on system on Linode Kubernetes Engine: upload user + garment images and render a synthesized try-on via CatVTON inference.
- Implemented a queue-based pipeline with Redis (job queue + pub/sub) and a GPU worker (Python/PyTorch) that pulls jobs, runs inference, uploads results to S3-compatible storage, and emits completion events.
- Delivered real-time UX with a Gateway WebSocket that routes per-user job-done events, enabling automatic frontend refresh without page reloads; used presigned S3 URLs for direct browser uploads.

The MyAnimeList Populator

[GitHub](#)

Node.js, Express.js, JavaScript, TypeScript, Angular, PostgreSQL, Docker, Google Cloud Run, Firebase Hosting

- Built and published an Angular SPA with a Dockerized Node.js/Express/PostgreSQL backend, deploying the API to Google Cloud Run (Cloud SQL) and hosting the frontend on Firebase Hosting for fast, reliable access.
- Engineered a Tinder-style swipe interface that empowers users to discover and curate their anime lists, adding and rating titles with a single fluid gesture.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C#, SQL

Frameworks and Libraries: React, Node.js, Express.js, Angular, .NET Core, FastAPI, PyTorch, SQLAlchemy

Technologies and Tools: PostgreSQL, MySQL, Redis, Docker, Kubernetes, AWS, GCP, Firebase, gRPC, Git